Does Political Reservation Affect Program Implementation? Evidence from India

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Abstract

Political reservation brings disadvantaged groups into elected leadership and can change policy choices and the distribution of public benefits, but less is known about its effects on the scale and implementation of public programs. These margins matter because they determine how many public assets are built and how many paid workdays are generated. I examine how reserving village council leadership for Scheduled Castes (SCs), Other Backward Classes (OBCs), and women affects the number of public works approved and shares of those works that are completed or abandoned. I construct a novel panel dataset from administrative records covering approximately 10 million public works across two council terms in Uttar Pradesh. Using two-way fixed effects, with a fuzzy regression discontinuity design as a complementary identification strategy, I find that SC reservation reduces the number of approved works by about 7 percent and raises the abandonment rate among approved works by 3.4 percent relative to its mean. SC reservation also reduces paid employment by 11 percent, equivalent to about 5.4 million fewer workdays across Uttar Pradesh each year. The effects are smaller under OBC reservation, while women's reservation has no consistent effect. The SC implementation gap is concentrated in community projects and narrows over the council term, consistent with learning.

JEL classification: D72, D73, H54, and O12.

Keywords: Political reservation, program implementation, NREGS, Gram Panchayats, India.

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1 Introduction

Many countries use political affirmative action by reserving elected offices for historically disadvantaged groups. India, the world's largest democracy, uses this policy to increase the representation of disadvantaged castes, tribes, and women. Such mandated representation is intended to give these groups greater influence over policy, and reserved leaders in India do shift policies and resources toward their own groups (Besley et al., 2004; Chattopadhyay and Duflo, 2004*b*; Pande, 2003). By changing who holds office, political reservation may affect not only which policies are chosen but also how effectively they are implemented. Understanding whether political reservation affects implementation is especially important in low- and middle-income countries, including India. Implementation failures are common in these settings and can waste public resources and prevent citizens from receiving the intended benefits (Lehne, Shapiro and Vanden Eynde, 2018; Muralidharan, Niehaus and Sukhtankar, 2016; Olken, 2007). For example, one-third of over 14,000 local projects started in Ghana were never completed (Williams, 2017), while only 31 percent of roughly 4,700 federal projects in Nigeria were fully completed (Rasul and Rogger, 2018).

This raises an important question: does political reservation affect program implementation? I examine two margins: the number of public works local governments undertake and the share they carry through to completion. I study these outcomes in India's National Rural Employment Guarantee Scheme (NREGS), the world's largest public employment program. NREGS guarantees rural households up to 100 days of paid work each year and finances local assets such as roads, irrigation, and water-conservation projects. I also examine how reservation affects the number of days of NREGS employment provided and the total wages paid to workers. Because village councils plan NREGS projects and implement a large share of them (Afridi, Iversen and Sharan, 2017; Sukhtankar, 2017), their performance matters for both household incomes and local infrastructure.

My analysis focuses on Uttar Pradesh, India's most populous state, whose population is comparable to that of the world's fifth-most-populous country. I study reservation of the *Sarpanch* position. The Sarpanch is the elected head of the Gram Panchayat (GP), a village council at the lowest tier of government in India. The Sarpanch plays a central role in decisions about local public goods and development programs, including NREGS (Afridi, Iversen and Sharan, 2017; Besley et al., 2004). Uttar Pradesh reserves Sarpanch positions by caste and gender. Caste reservations cover Scheduled Castes (SCs), Scheduled Tribes (STs), and Other Backward Classes (OBCs) and are assigned using population-based rules within blocks, which are subdistricts containing several

GPs.[1] At least one-third of positions, both within each caste category and overall, are also reserved for women (Chaturvedi, Das and Mahajan, 2025).

I identify the effect of political reservation on program implementation using two complementary research designs and administrative records covering more than ten million NREGS works in 49,527 GPs under the councils elected in 2015 and 2021. First, I exploit the rotation of reserved positions in a two-way fixed-effects design that compares the same GP across the 2015 and 2021 council terms. Because GPs switching status may follow different trends or past reservation may affect later outcomes, I complement this analysis with a fuzzy regression discontinuity design that exploits the population-based rule used to assign reservation in 2015. This design compares GPs in the same block with nearly identical population rankings but sharply different reservation probabilities. Similar estimates across the two designs strengthen the interpretation that the results reflect the causal effect of political reservation.

I find that SC reservation reduces both the number of works approved and how effectively those works are implemented. Compared with the same GP when its Sarpanch position is not reserved by caste, an SC-reserved GP has about 7 percent fewer approved works and 9 percent fewer completed works. Among approved works, the share completed falls by 0.8 percentage points, while the share abandoned rises by 0.7 percentage points, or 3.4 percent relative to the mean. The regression discontinuity design yields similar estimates, with 6.4 percent fewer works approved and a 1.2 percentage point increase in abandonment at the SC cutoff. The fixed-effects estimates for OBC reservation are about half as large, while the discontinuity estimates are insignificant. Women's reservation has small, generally insignificant effects across both designs, providing no consistent evidence that it weakens implementation. SC reservation also reduces the employment generated by these public works. SC-reserved GPs provide about 11 percent fewer days of paid NREGS employment, equivalent to approximately 5.4 million fewer person-days annually across Uttar Pradesh.

I next examine two patterns that may explain why SC-reserved GPs complete fewer approved works and abandon more of them. First, the gap is concentrated in community projects, particularly natural resource management works such as ponds and irrigation canals, and rural infrastructure works such as roads, drains, and public buildings. By contrast, the effects are small and only marginally significant for assets built for individual households, such as cattle sheds and farm

[1] I do not report effects of ST reservation because ST-reserved GPs constitute a negligible share of the sample. The regressions control for it.

ponds. This pattern may indicate that SC Sarpanches give greater priority to completing individual projects, which target poor and vulnerable households, many of which belong to SCs. Alternatively, caste-based resistance may make it harder for SC Sarpanches to secure cooperation from higher-level officials and residents of other castes. Such resistance would particularly hinder community projects, which require greater coordination. Second, the increase in abandonment under SC reservation is largest in the first year and falls by about two-thirds by the fifth year, consistent with SC Sarpanches improving implementation as they gain experience.

I make three contributions to the literature. First, I extend research on how political reservation affects public program implementation. Existing studies focus on outcomes different from those studied here. Das, Mukhopadhyay and Saroy (2023) find no average effect of OBC reservation in Rajasthan on NREGS employment per capita. Afridi, Iversen and Sharan (2017) find that women-reserved councils in Andhra Pradesh initially had more corruption and irregularities, although this gap narrowed with experience. Bose and Das (2018) study nearly 6,000 GPs in nine Uttar Pradesh districts and find that women-reserved GPs in poorer districts issue more job cards and undertake more water-harvesting projects, but do not differ in NREGS employment or works per rupee spent. I examine a central but largely overlooked dimension of program implementation: how many public works are approved during a GP's term and whether those works are ultimately completed or abandoned. To my knowledge, this is the first study to use detailed individual work records to estimate the effect of political reservation on both margins.

Second, I contribute to the mixed evidence on SC political reservation. SC reservation in state assemblies increased public-sector job quotas for SCs and STs but did not increase spending on SC welfare, reduce poverty, or improve literacy, employment, or village amenities (Chin and Prakash, 2011; Jensenius, 2015; Pande, 2003). Although greater SC representation increased SCs' access to NREGS employment relative to non-SCs, their relative outcomes in primary schooling, infant mortality, and access to subsidized food worsened (Bhattacharjee and Roy Chaudhuri, 2024). At the village level, some studies find that SC or ST reservation directs more public investment and household benefits toward disadvantaged groups (Bardhan, Mookherjee and Parra Torrado, 2010; Besley et al., 2004; Chattopadhyay and Duflo, 2004*a*), while others find weak effects on benefit allocation (Dunning and Nilekani, 2013). I extend the literature on SC political reservation to program implementation, showing that fewer works are approved under SC-reserved GPs and that these GPs complete a smaller share of approved works. The implementation gap, however, is largest early in the council term and falls by about two-thirds by the fifth year, consistent with learning

over time.

Third, I contribute to the literature on women's political reservation. Prior studies show that women's representation can improve health, education, and attention to women's policy priorities (Beaman et al., 2012; Bhalotra and Clots-Figueras, 2014; Chattopadhyay and Duflo, 2004*b*; Chaturvedi, Das and Mahajan, 2025; Clots-Figueras, 2012; Svaleryd, 2009). These effects can also vary by the caste of women leaders (Clots-Figueras, 2011). Under NREGS, women's reservation increases women's paid employment and shifts projects toward water control and harvesting without raising overall employment (Bose and Das, 2018; Deininger, Nagarajan and Singh, 2020). Evidence on women-led GPs is mixed. Women-led GPs perform similarly to male-led GPs on average (Ban and Rao, 2008) and exposure also reduces voter bias and improves women candidates' electoral prospects (Beaman et al., 2009). Other studies identify several limitations. Women-led GPs target fewer benefits toward disadvantaged households and complete fewer projects, such as roads and schools, that require coordination with higher-level officials (Bardhan, Mookherjee and Parra Torrado, 2010; Gajwani and Zhang, 2015). Women-reserved GPs also initially experience more NREGS corruption, but this gap narrows as leaders gain experience (Afridi, Iversen and Sharan, 2017). My findings add to this literature by showing that women's political reservation does not worsen NREGS implementation relative to unreserved councils, whether measured by project completion or local employment.

These findings should not be interpreted as evidence that SC leaders are inherently less capable. If the community-project gap reflects weaker cooperation, the constraint lies in caste barriers rather than leadership ability. If it instead reflects initial inexperience, its decline over the council term suggests that the cost may be temporary. Any short-run implementation costs should be weighed against the broader gains from representation. Reservation brings SCs into leadership positions they rarely win without it and has reduced dominant-caste residents' stated intentions to discriminate (Chauchard, 2014). In Bihar, it has also narrowed SC–non-SC gaps in public goods and household assets (Kumar and Sharan, 2026). My findings therefore motivate evaluating whether targeted administrative support, particularly early in a leader's term, can improve implementation while preserving the gains from political reservation.

The remainder of the paper is organized as follows. Section 2 provides institutional background. Section 3 describes the data and outcomes, and Section 4 presents the fixed-effects and fuzzy regression discontinuity designs and validity tests. Section 5 reports the main results and examines potential mechanisms. Section 6 concludes.

2 Background

2.1 The National Rural Employment Guarantee Scheme

The National Rural Employment Guarantee Act of 2005 legally guarantees rural households up to 100 days of paid employment annually when an adult household member volunteers to undertake unskilled manual work.[2] The program was introduced in three phases: the 200 poorest districts in 2006, another 130 districts in 2007, and all remaining rural districts by 2009 (Bose and Das, 2018). It has since become the world's largest workfare program, with cumulative spending of US$44.4 billion between 2005 and 2015 (Deininger, Nagarajan and Singh, 2020). Its two central objectives are improving local public infrastructure and providing employment when agricultural labor demand is low (Bose and Das, 2018). Projects are labor-intensive by design, with materials limited to 40 percent of project costs (Afridi, Iversen and Sharan, 2017). Eligible works include water conservation, drought proofing and flood mitigation, agricultural land development, irrigation, and rural roads (Afridi, Iversen and Sharan, 2017).

The selection and implementation of NREGS projects center on the Gram Panchayat (GP), an elected local council, led by Sarpanch (i.e., GP president), that may encompass more than one village. The Gram Sabha, comprising all registered voters within the GP's jurisdiction, meets regularly to identify and prioritize works based on local needs, in coordination with block- and district-level officials (Sukhtankar, 2017). The GP submits the ranked proposals to the block Programme Officer, who forwards them to the district Programme Officer for final technical and financial approval (Afridi, Iversen and Sharan, 2017). Before each financial year, officials approve a list of eligible works but do not determine when each work will begin. Instead, works are initiated as households request employment (Sukhtankar, 2017). The GP executes the works assigned to it, and the law requires at least half of all NREGS works, measured by cost, to be implemented through GPs (Afridi, Iversen and Sharan, 2017; Sukhtankar, 2017).

Final approval makes a work eligible for implementation but does not ensure that it will be started or completed. To carry out an approved work, the GP must open it, mobilize workers, maintain muster rolls, arrange materials, and resolve local disputes over land.[3] Besides delays and funding constraints, works can therefore remain unstarted or unfinished when this local planning and coordination fails, for example if a land dispute is left unresolved or materials are never

[2] See the National Rural Employment Guarantee Act, 2005.
[3] See Ministry of Rural Development, NREGA Operational Guidelines, 2013.

arranged.

In Uttar Pradesh, the Sarpanch is directly elected and has substantial authority over the GP's priorities and expenditures (Chaturvedi, Das and Mahajan, 2025). Evidence from elsewhere in India similarly shows that Sarpanches retain substantial discretion over local decisions despite requirements to consult villagers and ward members (Besley et al., 2004). Political reservation may therefore affect both which public works are selected and how effectively they are implemented.

2.2 Gram Panchayat elections and reservation

India's 73rd constitutional amendment, adopted in 1992 and effective in 1993, established elected rural councils at the village, block, and district levels. Each council serves a five-year term (Beaman et al., 2009; Chattopadhyay and Duflo, 2004*b*). The Gram Panchayat (GP) forms the lowest tier. It consists of elected ward members and a chairperson. In Uttar Pradesh, GP voters elect the chairperson directly (Besley et al., 2004; Chaturvedi, Das and Mahajan, 2025).[4] This paper studies the councils elected in 2015 and 2021. Uttar Pradesh reserves some Sarpanch positions for Scheduled Castes (SCs), Scheduled Tribes (STs), and Other Backward Classes (OBCs). Only members of the designated group may contest a caste-reserved position. Positions not reserved by caste are open to all groups (Besley et al., 2004; Pande, 2003). The caste categories are mutually exclusive, but gender reservation may overlap with any category.

For the 2015 election, caste reservations were assigned in three steps. Officials first determined the statewide number of positions for each group. The SC and ST totals were based on their population shares in the 2011 Census. The Census does not report OBC population, so officials used a 2015 caste survey and capped OBC reservation at 27 percent of all Sarpanch positions. Of the state's 59,163 positions, 336 were reserved for STs, 12,246 for SCs, and 15,974 for OBCs. Together, caste-reserved positions accounted for 48.3 percent of the total, below the 50 percent ceiling.

Officials next divided the statewide totals among the state's 821 blocks. ST positions were allocated according to each block's share of the state's ST population. For SCs, the initial quota was based on the block's SC population share, up to the statewide share of 20.7 percent. For example, a block with an SC population share of 15 percent initially reserved about 15 percent of its positions for SCs. A block with an SC share above 20.7 percent initially reserved about 20.7

[4]Uttar Pradesh officially calls the GP chairperson the *Pradhan*. I use *Sarpanch* throughout for consistency with the broader literature.

percent. This cap left some SC positions unassigned. Officials allocated them to blocks with SC shares above 20.7 percent, starting with those having the highest shares. OBC positions were based on each block's OBC population share but were capped at 27 percent of its Sarpanch positions.

Finally, officials assigned each block's quotas to particular GPs. They ranked eligible GPs from highest to lowest by their ST population shares and selected GPs until the block's ST quota was filled. After removing those GPs, they ranked the remaining GPs by their SC population shares and filled the SC quota. They then repeated the procedure for OBCs using the GPs that remained. Thus, officials first determined how many reserved positions each block would receive and then used population rankings to determine which GPs would receive them. Reservation was therefore assigned through a population-based rule rather than a lottery.[5]

At least one-third of the positions assigned to each caste category were reserved for women from that category. Within each category, the required number of women-reserved positions was assigned to the highest-ranked GPs under the same descending population-share ranking. Among GPs not reserved by caste, additional positions were reserved for women until women accounted for at least one-third of all Sarpanch positions in the block. These positions were assigned in descending order of each GP's population excluding SCs, STs, and OBCs. A position reserved jointly for SCs and women, for example, could be contested only by SC women, whereas a position reserved for women but not by caste was open to women of any caste (Chattopadhyay and Duflo, 2004*b*; Chaturvedi, Das and Mahajan, 2025).[6]

Reservation categories normally rotate so that, whenever possible, a GP does not receive the same category in consecutive elections. In 2015, however, the state began a new rotation cycle because a statewide reorganization had created 7,315 GPs and changed the boundaries of roughly 8,000 others. Reservation assignments from earlier elections were therefore ignored, a decision upheld by the Allahabad High Court. For the 2021 election, officials rotated reservations relative to 2015. They first assigned each category to GPs that had not received that category in 2015 and returned to previously reserved GPs only when necessary to fill the required quota. Both elections used the same population data: the 2011 Census for SCs and STs and the 2015 rapid survey for OBCs. The number of Sarpanch positions changed from 59,163 across 821 blocks in 2015 to 58,194 across 826 blocks in 2021.[7] I compare 49,527 GPs that can be matched across both elections.

[5]See the Allahabad High Court judgment in *Sant Ram Sharma v. State of U.P. and 5 Others* (October 9, 2015), which describes the official allocation procedure and population figures. A GP with fewer than two residents belonging to a group was ineligible for reservation for that group.
[6]See the Allahabad High Court judgment (2015) and the 2021 government order.
[7]See the Government of Uttar Pradesh's 2021 government order.

3 Data

I use two types of data. The first consists of work-level administrative records from NREGS's management information system (MIS).[8] The MIS lists every public work recorded in a GP since 2005, most of them under NREGS, and reports each work's code, status, type, wage and material payments, person-days generated, and start and completion dates. I restrict the sample to works approved for implementation during the two council terms studied: May 2016 through March 2021 for councils elected in 2015 and November 2021 through March 2026 for councils elected in 2021. An approved work is one proposed through the Gram Sabha, submitted by the GP, and cleared for implementation by block- and district-level officials. The Supplementary Appendix describes the sample construction in detail. The resulting analysis sample contains about 10 million works across 49,527 GPs.[9]

Table 1 reports the implementation status of approved works as of August 2026. Column 1 pools the two council terms, while columns 2 and 3 report them separately. Each approved work falls into one of four mutually exclusive categories. Works *not yet started* had been approved but remained open without any work having begun. *Ongoing* works are works that had begun but were not yet finished, including works that were paused. *Completed* works are works that were finished on the ground. *Abandoned* works are works that were closed without any work being done, so the planned asset was never built. Across both terms, 4.4 percent of approved works had not yet started, 6.2 percent were ongoing, 69.8 percent were completed, and 19.6 percent were abandoned.

In the analysis, I use the shares of approved works that are ongoing, completed, and abandoned as outcomes. Works not yet started remain in the sample but are not reported as a separate main outcome for two reasons. First, the four categories are mutually exclusive and exhaustive, so the share not yet started contains no additional information once the other three shares are reported. Second, these works remain open and may eventually begin. They are concentrated in the 2021 council term, making their status particularly sensitive to the shorter follow-up period.

Table 2 describes the size, composition, and outcomes of works by type. The average completed work received Rs. 84,000 in wage and material payments and generated 283 person-days of employment. From the 2015 to the 2021 councils, the share of works providing individual assets fell from 44.4 to 34.4 percent, while the share devoted to natural resource management rose from 33.1

[8] See Ministry of Rural Development, NREGS MIS Reports.

[9] I exclude 30 GPs with no recorded works in at least one term: 23 have works in only one term, and 7 have none in either. For consistency, the regression discontinuity analysis in Section 4.2 also excludes the 17 GPs with no works under the 2015 councils.

to 47.2 percent. Outcomes also differ sharply by type. Among approved individual assets, 85.3 percent were completed and 9.5 percent were abandoned. Abandonment reached 25.3 percent for natural resource management and 29.3 percent for rural infrastructure. It occurs in every detailed work category, ranging from 14.0 percent for works on individual land to 43.6 percent for rural sanitation.

The second source of data covers the 2015 and 2021 GP elections. I use the election records assembled by Narasimhan and Weaver (2024), which report the winner of each GP election, the winner's caste, education, age, and other characteristics, and whether the Sarpanch position was reserved. I supplement these records with data from Chaturvedi, Das and Mahajan (2025), which combine election information with population from the 2011 Census and the 2015 caste survey for Uttar Pradesh.

Table 3 summarizes the election data. Panel A reports the share of Sarpanch positions reserved for each group, and Panel B describes the winners. Table 4 reports how reservation status changed between the two elections. Two patterns stand out. First, reservation strongly affects who holds office. Women won 98.4 percent of positions reserved for women, compared with 18.0 percent of positions not reserved for women. Similarly, SC candidates won 98.1 percent of SC-reserved positions but only 4.9 percent of positions not reserved by caste. OBC candidates won 98.8 percent of OBC-reserved positions and 39.3 percent of positions not reserved by caste. Reservation therefore changes the caste of the elected Sarpanch less sharply for OBCs than for SCs because OBC candidates frequently win positions without caste reservation.

Second, reservation rotates extensively across GPs. Among the 49,527 GPs observed in both terms, reservation status changed for 57 percent under women's reservation, 41 percent under SC reservation, and 54 percent under OBC reservation. The numbers entering and leaving each category were nearly equal, and only 7 GPs remained SC-reserved and 18 remained OBC-reserved in both terms. This rotation provides the within-GP variation used in the main analysis, allowing me to compare the same GP under reserved and unreserved Sarpanch positions.

4 Empirical strategy

I use two complementary research designs. The main specification is a two-way fixed-effects design with GP and time fixed effects. It compares outcomes within the same GP across the 2015 and 2021 council terms as its reservation status changes. This design uses both terms and the full

sample. I complement it with a fuzzy regression discontinuity design that compares similar GPs that narrowly did and did not qualify for reservation in 2015.

4.1 Two-way fixed-effects design

The main analysis addresses two questions. First, does reservation change how many works are approved for implementation under a GP? Second, among these approved works, does reservation change the shares that are ongoing, completed, or abandoned? The first question uses one observation per GP and council term, while the second uses one observation per approved work.

For GP g in council term t, I estimate

$$N_{gt} = \alpha_g + \delta_t + \beta \,\text{Women}_{gt} + \theta_{SC}\,\text{SC}_{gt} + \theta_{OBC}\,\text{OBC}_{gt} + \theta_{ST}\,\text{ST}_{gt} + \varepsilon_{gt}, \tag{1}$$

where N_{gt} is the number of works per year approved for implementation under GP g during term t; Women_{gt}, SC_{gt}, OBC_{gt}, and ST_{gt} are indicators equal to one if the Sarpanch position of GP g was reserved for that group in term t; α_g are GP fixed effects; δ_t is a term fixed effect; and ε_{gt} is an error term.[10] Works are counted per year because the two terms' windows differ in length, at 59 and 53 months, and the term fixed effect accounts for other differences common to a term. I report each coefficient as a percentage of the mean for GP-terms not reserved by caste. Standard errors are clustered by GP.

GP fixed effects account for permanent differences across GPs, such as caste composition and long-standing administrative capacity. The term fixed effect accounts for changes that affect all GPs between the two terms. Suppose, for example, that a statewide funding cut caused fewer works to be approved for implementation under all GPs during the 2021 term. The term fixed effect accounts for this common decline. If the decline was larger in GPs that became SC-reserved than in GPs that remained unreserved by caste in both terms, the additional decline is attributed to SC reservation. The identifying assumption is that, without the change in SC reservation status, both groups would have experienced the same decline. The corresponding assumption applies to OBC and women's reservation.

The number of approved works does not show how those works were implemented. I therefore ask whether reservation also affects whether approved works are completed or abandoned. To do

[10] Each term's window opens six months after the newly elected Sarpanches took office so that works inherited from the previous Sarpanch's pipeline are not attributed to the new one. The election month of April 2021, which also carries placeholder start dates for works that never began, is excluded.

so, I estimate a second specification using individual approved works. For work w in GP g, assigned to council term $t(w)$ and with a start date recorded in financial year $y(w)$, I estimate

$$Y_w = \alpha_g + \delta_{y(w)} + \beta \, \text{Women}_{g\,t(w)} + \theta_{SC} \, \text{SC}_{g\,t(w)} + \theta_{OBC} \, \text{OBC}_{g\,t(w)} + \theta_{ST} \, \text{ST}_{g\,t(w)} + \varepsilon_w, \qquad (2)$$

where Y_w is one of the three outcomes defined in Section 3: Ongoing, Completed, or Abandoned. The specification includes GP fixed effects, α_g, and recorded start-year fixed effects, $\delta_{y(w)}$. Since the outcomes are binary, the coefficients measure changes in the corresponding shares of approved works and are reported in percentage points.

Equation (2) is estimated at the work level, with one observation for each approved work. This allows me to include fixed effects for each work's recorded start year rather than only for its council term. A term fixed effect treats works recorded in 2022 and 2025 alike, even though the former have had much longer to be completed or abandoned. Recorded start-year fixed effects account for average differences in follow-up time and program conditions across years.[11] Consider a GP that was not reserved by caste in 2015 but became SC-reserved in 2021. Its 2018 works belong to the unreserved term, while its 2023 works belong to the SC-reserved term. The start-year fixed effects account for average differences between the 2018 and 2023 work cohorts across all GPs before the additional change associated with SC reservation is estimated.

If completion fell or abandonment rose more in GPs that became SC-reserved than in GPs that were not SC-reserved in either term, the additional change is attributed to SC reservation. The estimate also compares GPs that left SC reservation with GPs that were not SC-reserved in either term. The identifying assumption is that, without these changes in SC reservation status, the groups would have experienced the same change in implementation. The corresponding assumption applies to OBC and women's reservation.[12] The specification uses all 10.1 million approved works in the 49,527 GPs observed in both terms.

The assignment rule further supports these comparisons. Both elections used the same predetermined population data, and changes in reservation status followed population rankings, block quotas, and previous reservation status rather than past work outcomes. All regressions control for ST reservation, but I do not report its coefficient because few GPs are reserved for STs.

[11] For works that began, the recorded year is the actual start year. For works that never began, the MIS retains the date entered when the work was approved, so the year identifies the approval cohort.

[12] The start-year fixed effects yield conservative estimates. Replacing them with council-term fixed effects produces somewhat larger effects of caste reservation on completion and abandonment and leaves the estimates for women's reservation essentially unchanged.

Three issues require attention when interpreting the coefficients. First, work status is observed in August 2026. Recorded start-year fixed effects account for average differences in follow-up time across work cohorts, but the outcomes measure each work's status in August 2026 rather than its eventual status. Second, reservation may change the types of projects a GP undertakes, and abandonment rates vary widely across project types. The estimated effect could therefore partly reflect a change in project composition. To examine this possibility, I estimate the work-level equation separately within each major work category. Third, each pooled coefficient combines GPs entering and leaving reservation. If these effects differ, the pooled estimate may conceal meaningful differences (de Chaisemartin and D'Haultfœuille, 2020). I therefore estimate the two effects separately, using GPs that were never reserved for that category as the comparison group.

There are two remaining concerns that may threaten identification in the two-way fixed-effects design. First, reservation is based on population rankings rather than random assignment. GP fixed effects account for permanent differences in caste composition, but they do not rule out the possibility that GPs with larger SC or OBC population shares changed differently between the two terms for reasons unrelated to reservation. Second, a GP's reservation status in 2021 depends partly on whether it was reserved in 2015. If the effects of the 2015 reservation continued into the next term, perhaps through unfinished works, outcomes in 2021 could reflect both current and previous reservation. Together, these concerns motivate a second research design that does not rely on comparisons across terms.

4.2 Fuzzy regression discontinuity design

I complement the fixed-effects analysis with a fuzzy regression discontinuity design for the councils elected in 2015. The design compares GPs in the same block and council term that have nearly identical population rankings but different predicted reservation statuses. Since it relies on neither changes across terms nor the rotation of reserved positions, it provides an independent test of the main results.

I apply this design only to the 2015 election. Reservation normally rotates relative to the previous election, so a GP's assignment depends partly on its earlier reservation status. The 2015 election was an exception. Following a statewide reorganization of GPs, reservations were assigned without regard to the 2010 assignments. A GP's position relative to the 2015 cutoff therefore depended only on its population measure. Rotation resumed in 2021, when each category was assigned first to GPs that had not held it in 2015. A discontinuity based on the 2021 ranking would

therefore mix the effect of current reservation with the influence of previous reservation status.

As described in Section 2.2, officials ranked eligible GPs within each block by the population measure relevant to each reservation category and assigned positions from the top of the ranking until the block's quota was filled. The lowest-ranked GP predicted to receive a reserved position and the highest-ranked GP predicted not to receive one have similar population measures but different predicted reservation statuses.

For clarity, I describe the design for one reservation category and suppress the category index. Let P_g denote the population measure used to rank GP g. For block b, I define the cutoff as the midpoint between the last GP predicted to be reserved and the first GP predicted not to be reserved:

$$c_b = \frac{P_b^{\text{last}} + P_b^{\text{first}}}{2}. \tag{3}$$

The running variable is the GP's distance from its block-specific cutoff:

$$X_g = \frac{P_g - c_b}{\sigma}, \tag{4}$$

where σ is the standard deviation of that ranking measure among all eligible Gram Panchayats in the state. Because each reservation category is ranked on a different measure, σ is computed separately for each, so that distance from the cutoff is expressed in comparable units. Higher-ranked GPs are predicted to be reserved, so predicted assignment is

$$A_g = \mathbf{1}\{X_g \geq 0\}. \tag{5}$$

Centering the running variable at each block's cutoff allows me to pool GPs across blocks. The RDD data cover 48,477 GPs in 810 of the state's 821 blocks. The sample is smaller than the main sample because reconstructing the assignment rule requires linking each GP to the 2015 State Election Commission records. This additional linkage is not required for the main specification. The estimation sample also differs across reservation categories because each category ranks a different set of eligible GPs and uses a different bandwidth.

I estimate separate designs for SC, OBC, and women's reservation. ST reservation is too rare to analyze separately, but I account for it when reconstructing the assignment process. The SC design ranks GPs remaining after the ST assignment by their SC population shares. The OBC design ranks GPs remaining after the ST and SC assignments by their OBC population shares.

For women's reservation within a caste category, GPs are ranked by that group's population share. GPs not reserved by caste are ranked by their population outside the SC, ST, and OBC groups.

Let Q_g denote actual reservation status. If the reconstructed assignment exactly reproduced the official allocation, Q_g would equal predicted assignment, A_g. In practice, the two do not always coincide.[13] Predicted assignment matches actual reservation for 99.5 percent of GPs for SC reservation, 99.0 percent for OBC reservation, and 97.9 percent for women's reservation. Within 0.1 standard deviations of the cutoff, the corresponding match rates are 95.5, 88.6, and 91.0 percent.

I therefore use predicted assignment as an instrument for actual reservation. The first-stage equation is

$$Q_g = \gamma A_g + \beta_1 X_g + \beta_2 A_g X_g + \eta_b + u_g, \tag{6}$$

where η_b denotes block fixed effects. The interaction between A_g and X_g allows the slope of the running variable to differ across the cutoff.

Figure 1 plots the share of GPs actually reserved against the running variable. The reserved share is close to zero below the cutoff and rises sharply at the threshold. At the bandwidths used for estimation, the first-stage discontinuity, γ, is 0.96 for SC reservation, 0.76 for OBC reservation, and 0.72 for women's reservation. Predicted assignment therefore produces a large change in actual reservation, although compliance with the rule is not perfect.

The first outcome is the number of works approved for implementation under the GP per year under the 2015 councils. The remaining outcomes are the shares of its approved works that were ongoing, completed, or abandoned. I estimate the following outcome equation:

$$Y_g = \tau Q_g + \pi_1 X_g + \pi_2 A_g X_g + \eta_b + \varepsilon_g, \tag{7}$$

where actual reservation, Q_g, is instrumented with predicted assignment, A_g, using equation (6). The terms $\pi_1 X_g$ and $\pi_2 A_g X_g$ allow the relationship between the outcome and the running variable to differ across the cutoff. Block fixed effects account for block-level differences and ensure that GPs are compared within the same block.

Under the standard continuity, exclusion, and monotonicity assumptions, τ identifies the local average effect of reservation for GPs whose actual reservation status changes at the predicted-assignment cutoff. Continuity requires that, without a change in reservation status, outcomes

[13] For SC and OBC reservation, the statutory formulas reproduce the official quotas in all 821 blocks. The remaining discrepancies concern which GPs receive those positions. One likely source is the mapping of 2011 Census population counts to GP boundaries that changed in 2014–15, since small differences can alter rankings near the cutoff.

would vary smoothly at the cutoff. Exclusion requires predicted assignment to affect the outcomes only through actual reservation. Monotonicity requires that crossing the cutoff cannot make a GP less likely to be reserved.

I estimate equation (7) on GPs within the selected bandwidth using a triangular kernel and cluster standard errors by block. For the number of approved works, each GP is weighted only by the kernel. Since larger GPs undertake more works, this regression controls for GP population in the 2011 Census, which was measured before reservation was assigned (Calonico et al., 2019). I report the estimate as a percentage of the average number of works just below the cutoff.

For the three implementation shares, I multiply the kernel weight by the number of approved works in the GP. This gives the same relative weight to works as a work-level regression, making the estimates comparable to equation (2). The two designs are not identical, however, because the discontinuity specification cannot include work start-year fixed effects.

The bandwidth determines which GPs enter the local comparison. Following Calonico, Cattaneo and Titiunik (2014), I use a mean-squared-error-optimal bandwidth, which balances the bias from a wider window against the imprecision from a narrower one. Since the optimal bandwidth is outcome specific, selecting it separately for every outcome would change the estimation sample across columns. I instead select one bandwidth for each reservation category using the abandonment rate and apply it to all outcomes, including the number of works. Every estimate for a given reservation category therefore uses the same range of the running variable. The selected bandwidths are 0.668 standard deviations for SC reservation, 0.331 for OBC reservation, and 0.318 for women's reservation.[14] I also report estimates using one-half and one-quarter of each selected bandwidth.

4.3 Validity of the discontinuity design

The discontinuity design requires that GPs just above and below the cutoff be comparable and that they could not precisely manipulate their position around the cutoff. Such manipulation is unlikely in this setting. The ranking measures come from the 2011 Census and the 2015 caste survey, both of which predate the reservation assignments. Each block's cutoff also depends on the rankings of all GPs in that block, limiting any single GP's ability to determine its position relative to the cutoff. Related studies similarly treat official population counts used to allocate government programs as difficult to manipulate (Asher and Novosad, 2020; Burlig and Preonas, 2024).

[14] Selecting the bandwidth using the completion rate instead gives 0.673 for SC reservation, 0.286 for OBC reservation, and 0.315 for women's reservation.

Figure 2 shows no unusual clustering of GPs immediately above the cutoff, where the assignment rule predicts reservation. In fact, fewer GPs appear just above the cutoff than just below it. This pattern could reflect sorting away from reservation, but such sorting is unlikely because the rankings are based on population data from the 2011 Census and the 2015 caste survey, both recorded before the 2015 assignments. Some clustering around zero also arises automatically from the cutoff construction. In each block, the cutoff is placed between the last GP predicted to be reserved and the first GP predicted not to be reserved, putting one GP on either side of zero. The estimated density difference is also sensitive to the bandwidth used.[15] Overall, I find no clear evidence of manipulation around the SC cutoff, which is central to the paper's main findings.

GPs just above and below the cutoff are also similar in characteristics measured before the 2015 councils took office (Table A1). These characteristics include population, local infrastructure recorded in the 2011 Census, and NREGS employment during 2012–2014. For SC reservation, where the main effects appear, none of the 16 differences is statistically significant. Across all three reservation categories, only 3 of the 48 differences are significant at the 10 percent level. Joint tests also fail to reject balance for every category. The most relevant difference is at the OBC cutoff, where GPs just above the threshold have about 6 percent fewer residents in total. Since larger GPs tend to have more works approved for implementation, the regressions for the number of works control for GP population.

The predetermined ranking measures, absence of excess mass above the cutoff, and balance in pre-assignment characteristics support the continuity assumption. The cutoff has no administrative role beyond assigning reservation, which supports the exclusion restriction. Crossing the SC cutoff, however, also lowers the chance that a seat is reserved for women, since women's seats within each caste category go to the highest-ranked GPs. Section 5.3 shows that the SC estimates hold when GPs reserved for women are dropped. The assignment rule and large positive first-stage discontinuities make violations of monotonicity unlikely.

[15] Using the same bandwidth on both sides, the estimated discontinuity is insignificant for SC reservation ($p = 0.47$). The corresponding p-values are 0.08 for OBC reservation and 0.05 for women's reservation. For women's reservation, the difference is concentrated among GPs not reserved by caste; no discontinuity appears among SC- or OBC-reserved GPs.

5 Results

5.1 The number of works

SC reservation reduces the number of works approved for implementation under a GP. Figure 3 presents estimates of equation (1), while Table A2 reports the corresponding coefficients. GPs average 22.4 approved works per year when their Sarpanch position is not reserved by caste. When the position is reserved for SCs, about 1.5 fewer works are approved per year, a decline of 6.9 percent. OBC reservation reduces the number of approved works by 3.6 percent, about half the SC effect. Women's reservation has no measurable effect, with an estimate equal to 0.4 percent of the mean and statistically insignificant. The SC and OBC estimates remain almost unchanged when block-by-term fixed effects replace the term fixed effect and absorb shocks shared by GPs in the same block and term. Both estimates also remain significant at the 1 percent level when standard errors are clustered by block (Table A2, columns 2 and 3).

The decline in approved works under caste reservation could arise at several points in the approval process. As described in Section 2.1, a work is proposed in the Gram Sabha, submitted by the GP, and approved for implementation only after review by block and district officials. Since the comparison holds the GP itself fixed, the decline reflects something that changes with its leadership, and three explanations stand out. First, the GP may submit fewer proposals. Residents may bring fewer works to the Gram Sabha, perhaps because they trust the new leadership less, or the GP itself may prepare fewer proposals. Second, more proposals may be rejected because they do not meet the program's rules. NREGS caps material costs at 40 percent of total costs and requires at least 60 percent of district spending to go to works related to agriculture and natural resources. Each work must also be technically feasible. A pond, for example, must have workable dimensions. Third, block officials or engineers may discriminate against SC or OBC Sarpanches and approve fewer of their proposals. The number of approved works, however, is only one part of the story. I next examine how effectively those works are implemented.

5.2 Implementation of approved works

Caste reservation also weakens the implementation of approved works. Table 5 reports estimates of equation (2) for three outcomes: the shares of approved works that are ongoing, completed, and abandoned. When a GP's Sarpanch position is reserved for SCs, its approved works are 0.76 percentage points less likely to be completed and 0.67 percentage points more likely to be abandoned.

Relative to the sample means of 69.8 and 19.6 percent, completion falls by 1.1 percent and abandonment rises by 3.4 percent. The gap does not reflect works that are still under way: the effect on the share of ongoing works is close to zero and statistically insignificant.[16] For OBC reservation, the estimates are about half as large.[17] Women's reservation does not affect abandonment. Its 0.25 percentage point lower completion rate is partly matched by a 0.12 percentage point higher share of ongoing works, consistent with a small delay rather than a failure to carry out works. The SC estimates are nearly unchanged with block-by-start-year fixed effects and with standard errors clustered by block (Table A4). Table A5 separates GPs that entered a reservation category in 2021 from those that left it. The SC result is driven mainly by GPs entering SC reservation, while the OBC result is driven mainly by GPs leaving OBC reservation.

Taken together, the results show that fewer works are approved under caste-reserved GPs and that, among approved works, a smaller share is completed and a larger share is abandoned. These differences are more pronounced under SC reservation than under OBC reservation. Both effects reduce the number of completed projects in the village. An SC-reserved council completes about 1.4 fewer works a year than the same GP when its position is not reserved by caste, roughly 9 percent fewer (Table 6, column 1).

I also measure the employment forgone, since providing guaranteed employment is one of the main objectives of large-scale programs like NREGS. I find that an SC-reserved council provides about 530 fewer person-days of employment a year, roughly 11 percent fewer, and pays about Rs. 107,000 less in wages (Table 6, columns 2 and 3). Employment falls by more than the number of completed works, consistent with the gap being concentrated in large community works, which employ more workers (Section 5.4). For OBC reservation, the employment effects are about a third as large, and women's reservation has no measurable effect. Across the roughly 10,000 SC-reserved GPs in each term, the forgone employment amounts to about 5.4 million person-days and Rs. 1.1 billion in wages a year. Thus, the implementation gap under SC reservation means not only fewer public assets but also less work and income for rural households.

[16] The remaining approved works have not yet started. SC reservation raises their share by 0.14 percentage points (standard error 0.04), so slower starts explain only a small part of the gap.

[17] The SC effect is similar on seats also reserved for women. Of the 20,451 SC-reserved GP-terms, 7,236 are also reserved for women. SC reservation raises abandonment by 0.6 percentage points without women's reservation and by 0.8 percentage points with it, and lowers completion by 0.8 and 0.6 percentage points, respectively (Table A3). Neither difference is statistically significant.

5.3 Evidence from the regression discontinuity design

I complement the main estimates with the fuzzy regression discontinuity design based on the 2015 assignment rule, re-estimating the effects of reservation on both the number of works approved for implementation and the implementation of those approved works. The design compares GPs just above and below the assignment cutoff, which have similar population shares but different chances of being reserved. Since it uses only the 2015 assignment, it does not depend on how reservation rotated between the two terms.

Starting with the number of approved works, the discontinuity design confirms the decline under SC reservation. Panel A of Figure 4 reports the estimates, and Panel B plots the number of works approved per year against the running variable. At the SC cutoff, reservation reduces the number of works approved for implementation under a GP by 1.5 per year, or 6.4 percent of the mean just below the cutoff (23.2 works per year). The estimate grows somewhat as the bandwidth narrows, to 7.2 percent at half and 8.5 percent at a quarter of the MSE-optimal bandwidth, and it is 6.4 percent when the regression also controls for the number of villages, GP area, and NREGS person-days in 2012–2014 (Table A6). All of these SC estimates are significant at the 1 percent level. The OBC estimates are small and statistically insignificant at every bandwidth. For women's reservation, the estimate suggests a decline of about 4 percent, marginally significant at the MSE-optimal bandwidth and significant with the extra controls, but it fades at narrower bandwidths and does not appear in the fixed-effects estimates.

Turning to implementation, Table 7 shows the same pattern as the fixed-effects estimates, although the effects are somewhat larger. SC reservation lowers the share of approved works completed by 1.2 percentage points, from a mean of 78.0 percent just below the cutoff, and raises the share abandoned by 1.2 percentage points, from a mean of 20.2 percent. The share of ongoing works does not change. The effects grow slightly as the bandwidth narrows, to 1.4 and 1.5 percentage points at a quarter of the bandwidth, and Figure 5 shows the corresponding jumps at the cutoff. The SC estimates also hold when the GPs closest to the cutoff, where any sorting would show up, are dropped: excluding GPs within 0.02 or 0.05 standard deviations of the cutoff leaves them between 0.9 and 1.1 percentage points, all statistically significant (Table A7). For OBC and women's reservation, the completion and abandonment estimates are smaller, none is statistically significant at any bandwidth, and the jumps in the figure are small.

The assignment rule also links women's reservation to caste reservation, since within each caste

category the seats for women go to the highest-ranked GPs. As a result, SC reservation lowers the chance that a seat is also reserved for women by 21 percentage points at the cutoff, where about a quarter of the GPs just below are reserved for women (Table A8, column 1). This link does not drive the SC estimates. When all GPs reserved for women are dropped, SC reservation still lowers the share of approved works completed by 1.0 percentage point, raises the share abandoned by 1.0 percentage point, and reduces the number of approved works by 7.7 percent, all significant at the 1 percent level (Panel B). The OBC estimates remain small and statistically insignificant.

The discontinuity estimates support the main results, especially for SC reservation. The estimated decline in approved works is nearly identical in the two designs, at 6.4 percent in the discontinuity design and 6.9 percent in the fixed-effects design. The implementation effects are somewhat larger in the discontinuity design, but the two designs identify different effects: the discontinuity design estimates a local effect for GPs near the assignment cutoff, whereas the fixed-effects design averages across all GPs whose reservation status changed. Nevertheless, every fixed-effects implementation estimate falls within the 95 percent confidence interval of the corresponding discontinuity estimate. Since the two designs use different sources of variation, their similar results make it unlikely that the fixed-effects estimates reflect different trends across GPs or carryover effects from earlier reservation.

5.4 Possible mechanisms

The two designs agree that SC reservation lowers completion and raises abandonment among approved works. This gap is notable because fewer works are also approved for implementation under SC-reserved GPs, which might otherwise ease implementation constraints. Several mechanisms could explain the gap. SC Sarpanches may have greater difficulty mobilizing workers, arranging materials, or resolving disputes over land and project sites. Households may also be less willing to request employment or participate if they have less trust in the leadership. Caste-based resistance may further limit cooperation with SC leaders. SC leaders may also give priority to works that benefit their own constituents, such as assets for poor households. Finally, newly elected SC Sarpanches may initially be less familiar with administrative procedures. The data do not allow me to test each explanation directly. Instead, I examine two of their implications. If coordination, cooperation, or leaders' priorities drive the gap, it should be larger for community projects than for individual assets. If inexperience is the problem, the gap should narrow as Sarpanches gain experience.

Priority of SC Sarpanch

The implementation gap under SC reservation is concentrated in community projects rather than in projects for individual households. Table 8 reports separate estimates by work category for the shares of approved works that were completed and abandoned. For natural resource management projects on common land, SC reservation lowers the share of approved works completed by 1.4 percentage points and raises the share abandoned by 1.0 percentage point. The effects are larger for rural infrastructure, including roads, drains, and public buildings, where completion falls by 1.8 percentage points and abandonment rises by 1.7 percentage points. For individual assets built on household land, both effects are small, at 0.3 and 0.2 percentage points, and only marginally significant. The overall implementation gap therefore arises primarily in broadly shared projects that require greater coordination and administrative effort. One possible explanation is that individual assets largely benefit households below the poverty line, many of which belong to SCs, giving SC leaders stronger reasons to complete them.

This distinction is consistent with earlier evidence that political reservation can produce different outcomes for targeted and broadly shared goods. Pande (2003) finds that SC reservation increases policies targeted directly toward SCs, particularly public-sector job quotas, without a comparable increase in general spending. Bhattacharjee and Roy Chaudhuri (2024) similarly find that greater SC representation improves access to targetable workfare employment but is associated with worse broad-based outcomes such as schooling, health, and subsidized food.

Inexperience

A possible explanation for the implementation gap is that newly elected SC Sarpanches are initially less familiar with administrative procedures. If so, the gap should be largest early in the council term and narrow as they gain experience. Table 9 tests this prediction by interacting SC reservation with the number of years since the Sarpanch took office. In the first year, SC reservation increases abandonment by 0.9 percentage points, but the effect declines by 0.15 percentage points with each additional year. By the fifth year, roughly two-thirds of the initial gap has disappeared. This decline is consistent with SC Sarpanches learning administrative procedures as they gain experience.

6 Conclusion

Political inclusion cannot deliver its intended benefits unless public programs are implemented effectively. I examine whether political reservation affects how effectively public works are implemented at the Gram Panchayat level. I combine administrative records on about 10 million public works with election and reservation data for the councils elected in Uttar Pradesh in 2015 and 2021. The rotation of reserved positions allows me to compare implementation within the same GP across reserved and unreserved council terms using a fixed-effects design. I complement this analysis with a fuzzy regression discontinuity design based on the 2015 assignment rule.

Across both research designs, SC reservation reduces both the number of works approved for implementation and the share carried through to completion. About 7 percent fewer works are approved under SC-reserved GPs than under the same GP when its position is not reserved by caste. These GPs complete about 9 percent fewer works and generate about 11 percent fewer person-days of employment. Among approved works, fewer are started and completed, while more are abandoned. These differences are concentrated in community projects, particularly natural resource management and rural infrastructure, and are much smaller for public works that benefit marginalized households. The implementation gap is largest early in the council term and narrows over time, consistent with a transition cost that declines as SC-reserved councils gain administrative experience. The estimates for OBC reservation are smaller, while women's reservation has little or no effect on public-work implementation. Across the roughly 10,000 SC-reserved GPs in each term, this gap amounts to about 5.4 million person-days of forgone employment each year.

I do not interpret these findings as evidence that political reservation reduces welfare overall. Reservation can improve welfare by bringing disadvantaged groups into political office, directing resources toward marginalized households, and reducing caste-based discrimination. Nor does the implementation gap imply that SC leaders are inherently less capable. Instead, the findings identify a distinct challenge: political inclusion alone may not ensure that approved public works are effectively implemented.

Implementing public works is a central responsibility of village council leaders. The results therefore motivate greater attention to program implementation when designing policies that accompany political reservation. An important question for future research is whether targeted administrative support, particularly early in the terms of leaders elected to caste-reserved positions, can reduce implementation gaps while preserving the broader gains from political inclusion.

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Tables and Figures

Table 1. Status of Approved Works ↩

	(1) All	(2) 2015 councils	(3) 2021 councils
Percent of approved works			
Not yet started	4.4	0.1	9.0
Ongoing	6.2	1.8	10.9
Completed	69.8	77.6	61.4
Abandoned	19.6	20.6	18.6
Number of approved works	10,149,181	5,272,065	4,877,116
Number of GPs	49,527	49,527	49,527

Notes: The table shows the status of approved NREGS works in the 49,527 GPs observed in both council terms, as recorded in August 2026. Column 2 covers works dated May 2016 to March 2021, under the councils elected in 2015. Column 3 covers works dated November 2021 to March 2026, under the councils elected in 2021. Works not yet started have not begun. Ongoing works have begun but are not finished, and include works that were halted. Completed works are finished. Abandoned works were closed without being carried out. Works under the 2015 councils have had more time, so few of them are still not yet started or ongoing.

Table 2. Works by Type ↩

	Work composition (%)		Outcomes among approved works (%)		Means among completed works	
	2015 (1)	2021 (2)	Completed (3)	Abandoned (4)	Payments (Rs. '000) (5)	Employment (person-days) (6)
Panel A. Master category						
A. Natural resource management	33.1	47.2	59.8	25.3	118.4	526
B. Individual assets	44.4	34.4	85.3	9.5	28.1	111
C. Self-help group infrastructure	0.0	0.0	40.1	48.3	246.7	177
D. Rural infrastructure	22.5	18.4	57.5	29.3	185.1	315
Panel B. Work category						
Works on individual land	51.1	39.5	80.4	14.0	25.4	101
Land development	17.2	24.7	65.1	23.5	116.3	533
Rural connectivity	8.1	9.0	59.5	20.6	240.7	273
Drought proofing	5.8	7.6	42.7	28.5	28.9	109
Water conservation and harvesting	5.0	5.6	59.0	27.4	238.3	1,082
Micro irrigation	2.9	5.3	69.3	21.6	112.4	432
Flood control and protection	3.8	4.0	67.7	22.8	150.0	560
Rural sanitation	3.6	0.5	52.1	43.6	27.9	86
Anganwadi and other infrastructure	1.3	2.5	52.0	28.5	439.1	325
Renovation of water bodies	1.2	1.4	62.2	22.7	245.6	1,059
Other	0.2	0.0	75.4	22.2	83.5	219
All works	100.0	100.0	69.8	19.6	84.0	283

Notes: The sample covers NREGS works in the 49,527 GPs observed in both council terms. Columns 1 and 2 report the distribution of all works, including new and deleted ones, across categories under the councils elected in 2015 and 2021, respectively. Columns 3 to 6 pool the two terms. Columns 3 and 4 report, within each category, the percentage of approved works that were completed (marked Completed or Physically Completed in the MIS) and the percentage that were abandoned (marked Closed). Column 5 reports mean wage and material payments per completed work, in thousands of rupees, and column 6 reports mean person-days of employment per completed work. Panel A classifies works by MIS master category. Individual assets are assets for vulnerable households on their own land, and self-help group infrastructure is shared infrastructure for self-help groups. Panel B uses the MIS work categories. "Other" combines other works, rural drinking water, fisheries, playgrounds, coastal-area projects, and Bharat Nirman Sewa Kendras. The two classifications are not nested: a work category may contain works from several master categories.

Table 3. Seat Reservation and Elected Sarpanches ↩

	(1) All	(2) 2015 councils	(3) 2021 councils
Panel A. Seat reservation (percent of GPs)			
Reserved for women	33.5	33.4	33.6
Reserved for SC	20.6	20.8	20.5
Reserved for OBC	27.0	27.2	26.9
Reserved for ST	0.5	0.5	0.5
Not reserved by caste	51.8	51.5	52.1
Panel B. Elected Sarpanch			
Woman (percent)	44.9	43.6	46.3
On seats reserved for women	98.4	98.3	98.5
On seats not reserved for women	18.0	16.2	19.8
SC (percent)	23.2	22.3	24.1
On seats reserved for SC	98.1	98.1	98.1
On seats not reserved by caste	4.9	2.6	7.2
OBC (percent)	47.3	40.6	54.0
On seats reserved for OBC	98.8	98.6	99.0
On seats not reserved by caste	39.3	26.3	52.1
Age (years)	41.4	41.1	41.8
Education (percent)			
Below high school	58.6	63.2	53.7
High school or intermediate	25.9	22.9	29.0
Graduate or above	15.5	13.8	17.3
Number of GP-terms	99,054	49,527	49,527

Notes: Each observation is a GP-term, and the sample contains the 49,527 GPs observed in both council terms. Column 1 pools both terms, while columns 2 and 3 correspond to the councils elected in 2015 and 2021, respectively. Panel A reports the percentage of GPs whose Sarpanch position was reserved for each group. The caste categories sum to 100 percent, subject to rounding; reservation for women may overlap with any caste category. Panel B reports the characteristics of the elected Sarpanch. Below high school includes illiterate Sarpanches and those with primary or junior-high education; high school or intermediate includes diploma holders; and graduate or above includes master's and doctoral degrees. Education shares exclude Sarpanches with no recorded education, who account for 4.8 percent of GPs in 2021.

Table 4. Reservation across the Two Council Terms ↩

	(1) Women	(2) SC	(3) OBC	(4) ST
Never reserved	18,845	29,083	22,781	49,024
Gained reservation in 2021	14,121	10,137	13,286	258
Lost reservation in 2021	14,055	10,300	13,442	243
Reserved in both terms	2,506	7	18	2
GPs observed in both terms	49,527	49,527	49,527	49,527

Notes: The table classifies the GPs observed in both council terms by whether their Sarpanch position was reserved for each group in 2015 and 2021. *Gained reservation in 2021* denotes GPs not reserved in 2015 but reserved in 2021, and *lost reservation in 2021* denotes GPs reserved in 2015 but not in 2021. Each column sums to the number of GPs observed in both terms.

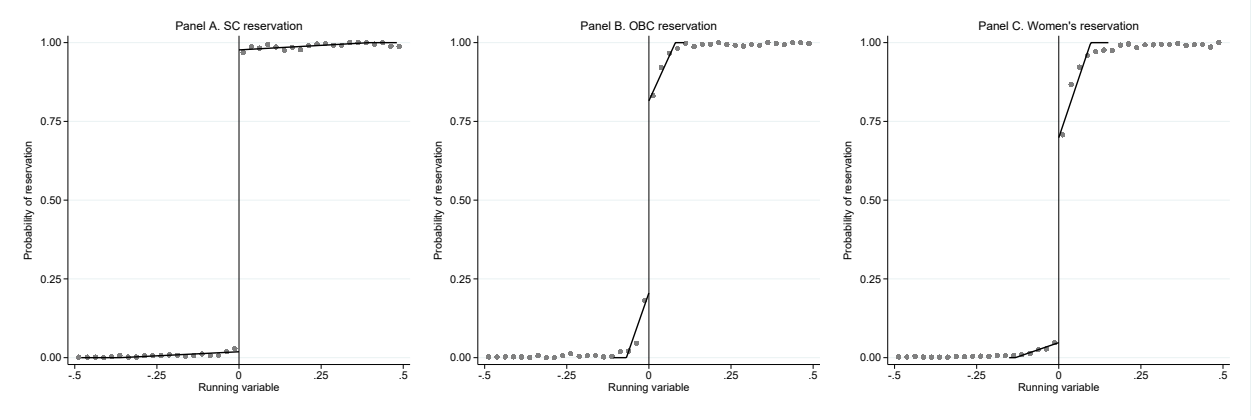


Figure 1. First Stage: Actual Reservation at the Cutoff ↩

Notes: Each panel shows how the share of Gram Panchayats reserved for the indicated group changes around the assignment cutoff. The dots report average reservation rates for groups of GPs with similar values of the running variable. The lines are fitted separately on each side of the cutoff and give greater weight to GPs closer to it. The figure is descriptive and does not account for the block fixed effects included in equation (6).

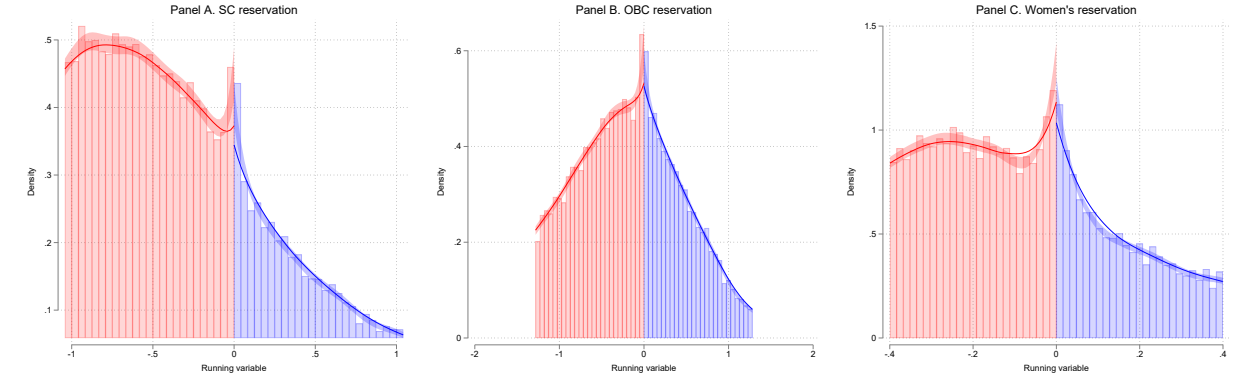


Figure 2. Density of the Running Variable at the Cutoff ↩

Notes: Each panel plots the estimated density of the running variable on either side of the cutoff, with 95 percent confidence bands, following Cattaneo, Jansson and Ma (2020). Bars show the distribution of Gram Panchayats. The estimated discontinuity is negative in all three cases, with p-values of 0.07 for SC, 0.09 for OBC, and 0.02 for women's reservation.

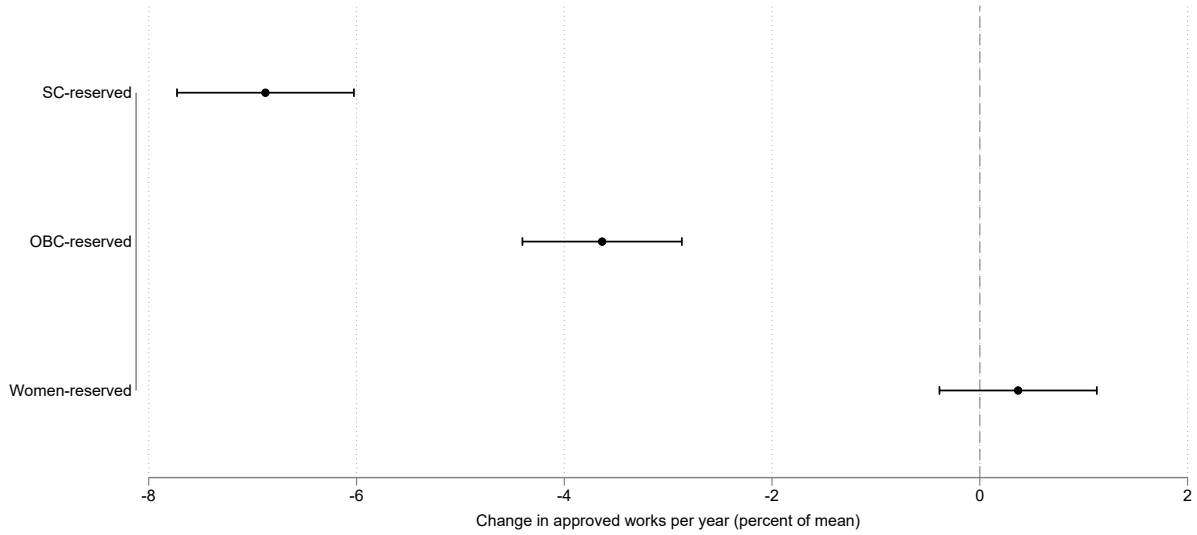


Figure 3. Effect of Reservation on the Number of Approved Works ↩

Notes: The figure shows the change in the number of works approved per year in a GP during a council term when its Sarpanch position was reserved, as a percent of the mean for GP-terms not reserved by caste (22.4 works per year). For SC and OBC reservation, the comparison is with terms in which the position was not reserved by caste; for women's reservation, with terms in which it was not reserved for women. Estimates are from equation (1), with GP and term fixed effects; bars show 95 percent confidence intervals, with standard errors clustered by GP.

Table 5. Effect of Reservation on Work Outcomes ↩

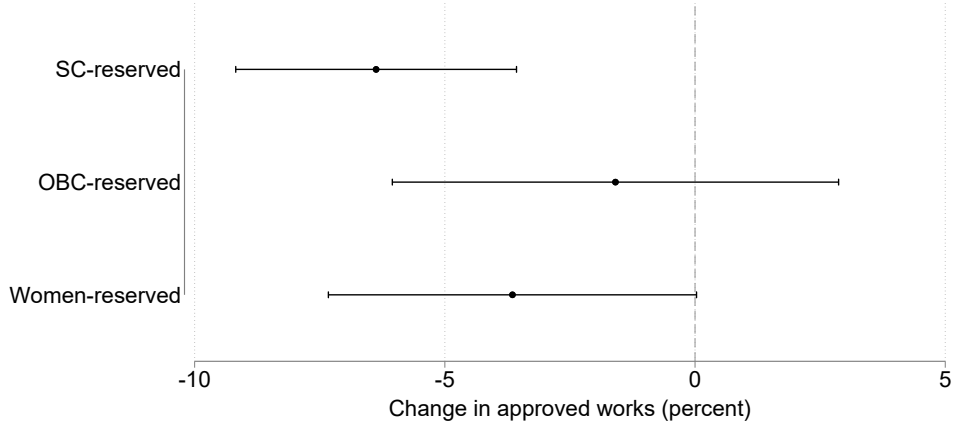
	(1) Ongoing	(2) Completed	(3) Abandoned
Women-reserved	0.0012***	-0.0025**	0.0008
	(0.0004)	(0.0010)	(0.0009)
OBC-reserved	0.0001	-0.0043***	0.0038***
	(0.0005)	(0.0011)	(0.0009)
SC-reserved	-0.0004	-0.0076***	0.0067***
	(0.0005)	(0.0012)	(0.0010)
Mean of outcome	0.062	0.698	0.196
Observations	10,149,181	10,149,181	10,149,181

Notes: The table reports estimates of equation (2). Each column is a separate regression of the indicated outcome on the reservation indicators, with GP and start-year fixed effects. ST reservation is included as a control but not reported. Standard errors, in parentheses, are clustered by GP. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

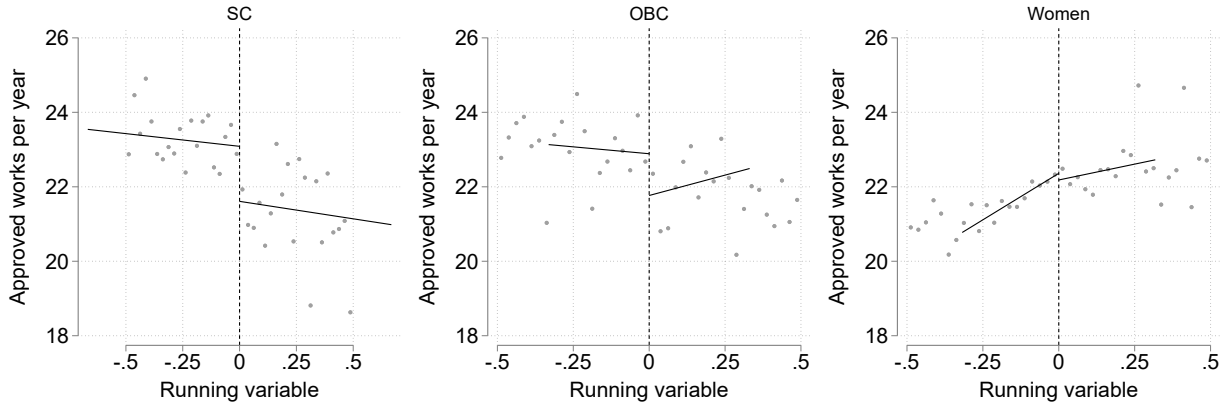
Table 6. Effect of Reservation on Completed Works, Employment, and Wages per Year ↩

	(1) Completed works	(2) Person-days (thousands)	(3) Wages paid (Rs. thousands)
Women-reserved	0.11	0.04	4.15
	(0.07)	(0.03)	(5.70)
OBC-reserved	-0.81***	-0.16***	-29.64***
	(0.07)	(0.03)	(5.88)
SC-reserved	-1.39***	-0.53***	-106.55***
	(0.08)	(0.03)	(6.49)
Mean, not reserved by caste	15.65	4.87	1020.32
GP-terms	99,054	99,054	99,054

Notes: Estimates are from equation (1), with GP and term fixed effects. All outcomes are per year. Column 1 is the number of works completed. Column 2 is the person-days of employment on all approved works, in thousands. Column 3 is the wages paid for that employment, in thousands of rupees. The mean row is the average for GPs in terms when the Sarpanch position was not reserved by caste. ST reservation is included as a control but not reported. Standard errors, in parentheses, are clustered by GP. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



Panel A. Effect of reservation on approved works (percent change)



Panel B. Approved works per year, by distance from the cutoff

Figure 4. Effect of Reservation on the Number of Approved Works per Year: Regression Discontinuity ↩

Notes: Panel A reports fuzzy regression discontinuity estimates of the effect of reservation on the number of works approved per year in a GP under the 2015 councils, as percent changes from the mean just below the cutoff, with 95 percent confidence intervals. Each regression uses the category's MSE-optimal bandwidth and block fixed effects, weights each GP only by the triangular kernel, controls for GP population, and clusters standard errors by block. Panel B plots approved works per year against the running variable, with binned means and local linear fits, and omits the block fixed effects and population control.

Table 7. Effect of Reservation on Work Outcomes: Regression Discontinuity Estimates ↩

	(1) Ongoing	(2) Completed	(3) Abandoned
Panel A. MSE-optimal bandwidth			
SC-reserved	-0.0003	-0.0118***	0.0119***
	(0.0007)	(0.0034)	(0.0034)
GPs	20,584	20,584	20,584
OBC-reserved	0.0004	-0.0040	0.0034
	(0.0011)	(0.0050)	(0.0048)
GPs	11,978	11,978	11,978
Women-reserved	0.0007	-0.0057	0.0050
	(0.0008)	(0.0039)	(0.0037)
GPs	22,450	22,450	22,450
Panel B. Half the bandwidth			
SC-reserved	-0.0009	-0.0134***	0.0141***
	(0.0009)	(0.0044)	(0.0043)
GPs	10,654	10,654	10,654
OBC-reserved	0.0031*	-0.0054	0.0023
	(0.0018)	(0.0080)	(0.0076)
GPs	6,513	6,513	6,513
Women-reserved	0.0004	-0.0045	0.0042
	(0.0012)	(0.0058)	(0.0055)
GPs	12,258	12,258	12,258
Panel C. One-quarter of the bandwidth			
SC-reserved	-0.0015	-0.0139**	0.0154***
	(0.0012)	(0.0058)	(0.0058)
GPs	5,569	5,569	5,569
OBC-reserved	0.0043	-0.0055	0.0018
	(0.0029)	(0.0121)	(0.0115)
GPs	3,538	3,538	3,538
Women-reserved	0.0007	-0.0044	0.0037
	(0.0017)	(0.0088)	(0.0084)
GPs	6,884	6,884	6,884
Mean below cutoff, SC-reserved	0.017	0.780	0.202
Mean below cutoff, OBC-reserved	0.018	0.778	0.204
Mean below cutoff, Women-reserved	0.017	0.771	0.211

Notes: The table reports fuzzy regression discontinuity estimates of equation (7) for the councils elected in 2015, instrumenting actual reservation with predicted assignment. The outcomes are the shares of a GP's approved works that are ongoing, completed, or abandoned. Each regression uses a triangular kernel and block fixed effects and weights each GP by its number of approved works. The MSE-optimal bandwidth (Calonico, Cattaneo and Titiunik, 2014) is selected once for each category using the abandonment rate: 0.668 standard deviations for SC, 0.331 for OBC, and 0.318 for women's reservation. Panels B and C use half and a quarter of it. Means below the cutoff are taken at the MSE-optimal bandwidth. Standard errors, clustered by block, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

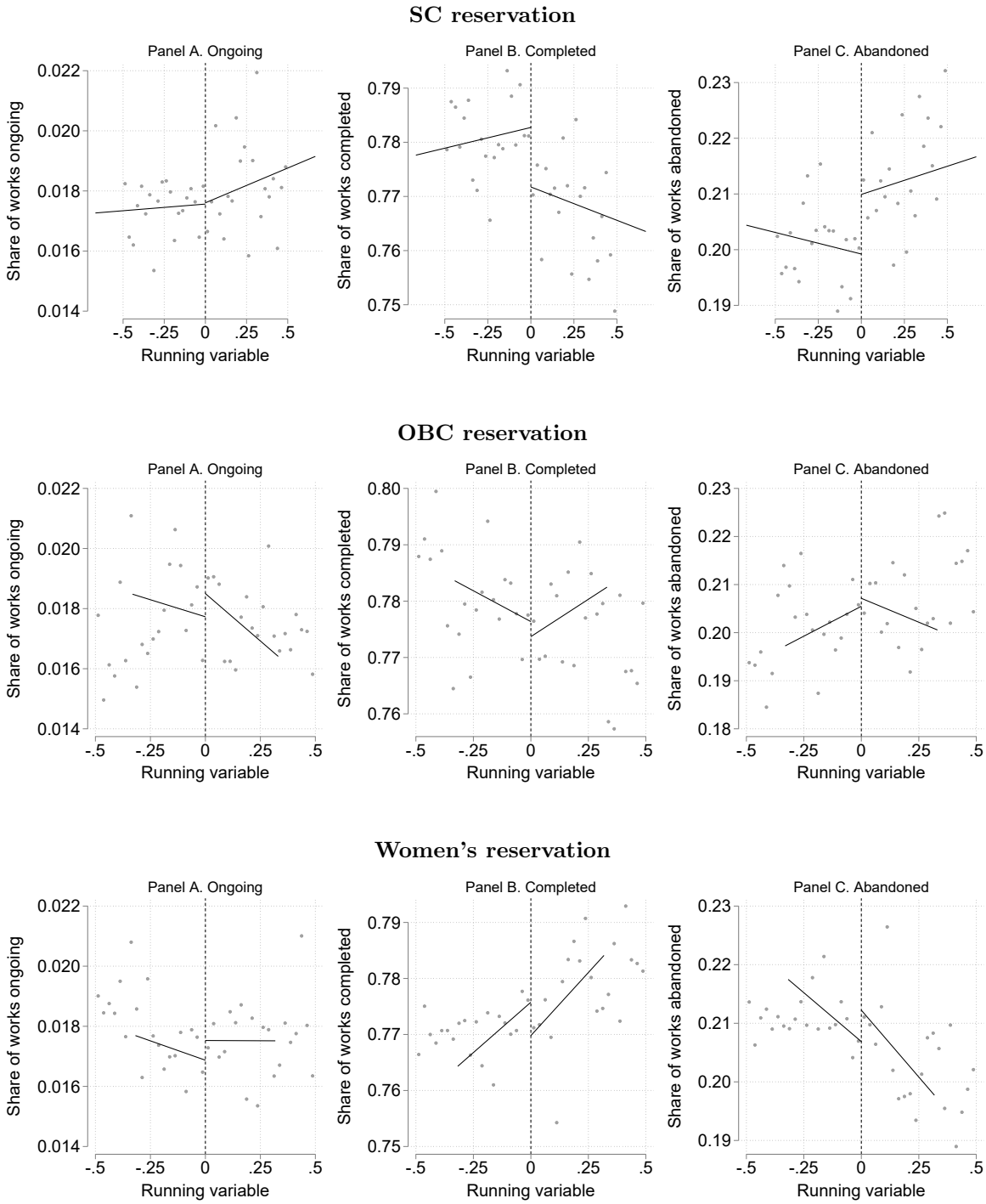


Figure 5. Effect of Reservation on Work Outcomes: Regression Discontinuity ↩

Notes: The figure covers the councils elected in 2015, with one row for each reservation category. Panels A to C plot the shares of approved works that are ongoing, completed, and abandoned against the running variable, which measures a GP's standardized distance from its block's reservation cutoff. Each dot is the mean outcome within a bin of 0.025 standard deviations, weighted by the number of approved works. Only GPs within 0.5 standard deviations of the cutoff are shown. The lines are local linear fits estimated separately on each side of the cutoff with a triangular kernel, within each category's MSE-optimal bandwidth: 0.668 standard deviations for SC, 0.331 for OBC, and 0.318 for women's reservation. The gap between the two lines at the cutoff shows the jump in the outcome. Unlike the regressions, the figure does not include block fixed effects.

Table 8. Effect of Reservation by Type of Work ↩

	Completed			Abandoned		
	(1) NRM	(2) Rural infra.	(3) Indiv. assets	(4) NRM	(5) Rural infra.	(6) Indiv. assets
Women-reserved	-0.000	-0.001	-0.001	0.000	-0.001	0.001
	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)	(0.001)
OBC-reserved	-0.003**	0.000	-0.005***	0.002*	-0.000	0.004***
	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)	(0.001)
SC-reserved	-0.014***	-0.018***	-0.003*	0.010***	0.017***	0.002*
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
Mean of outcome	0.598	0.575	0.853	0.253	0.293	0.095
Observations	3,971,469	2,005,742	4,168,852	3,971,469	2,005,742	4,168,852

Notes: Each column is a separate estimate of equation (2), with GP and start-year fixed effects, for approved works in the indicated MIS master category. Columns 1 to 3 report the share completed and columns 4 to 6 the share abandoned. NRM is natural resource management (category A: water and land works on common land), rural infrastructure is category D (roads, drains, and buildings), and individual assets are category B (assets for vulnerable households on their own land). Self-help group infrastructure (category C) is omitted because it contains few works. ST reservation is included as a control but not reported. Standard errors, in parentheses, are clustered by GP. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9. The SC Effect on Abandonment over the Sarpanch's Term ↩

	(1) Abandoned
SC-reserved	0.0094***
	(0.0014)
SC-reserved × years in term	-0.0015***
	(0.0006)
Mean of outcome	0.196
Observations	10,149,181

Notes: The outcome is whether an approved work was abandoned. The regression is equation (2) with an added interaction between SC reservation and years in term, which counts the full financial years since the councils took office. The SC-reserved coefficient is the effect at the start of the term, and the interaction is how much that effect changes with each additional year. Women's, OBC, and ST reservation are included as controls but not reported. Standard errors, in parentheses, are clustered by GP. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix

Table A1. Balance of GP Characteristics at the Cutoff ↩

	(1) SC	(2) OBC	(3) Women	(4) Mean
Panel A. 2011 Census				
Total population in the GP	8.2	-147.9**	-0.3	2,603.9
	(41.3)	(64.8)	(31.8)	
Villages in the GP	0.012	0.032	-0.029	1.651
	(0.030)	(0.037)	(0.029)	
Area (hectares)	0.6	-20.3*	10.7	383.2
	(10.3)	(12.4)	(8.2)	
Women per 1,000 men	1.0	-0.5	-0.3	924.2
	(1.6)	(2.3)	(1.5)	
Villages with a primary school	-0.011	-0.011	-0.003	0.812
	(0.009)	(0.011)	(0.008)	
Villages with a maternity centre	-0.008	-0.001	0.004	0.080
	(0.006)	(0.008)	(0.006)	
Villages with an anganwadi	-0.012	-0.006	0.004	0.845
	(0.009)	(0.011)	(0.008)	
Villages with a ration shop	-0.003	0.016	-0.000	0.752
	(0.011)	(0.013)	(0.010)	
Villages with tap water	0.003	-0.012	-0.018**	0.184
	(0.010)	(0.013)	(0.009)	
Villages with an all-weather road	0.010	-0.003	-0.007	0.716
	(0.011)	(0.014)	(0.010)	
Villages with power for homes	0.004	0.009	0.000	0.913
	(0.007)	(0.008)	(0.006)	
Villages with mobile coverage	0.011	0.014	-0.001	0.752
	(0.011)	(0.013)	(0.010)	
Share of area irrigated	-0.006	0.006	0.005	0.872
	(0.004)	(0.006)	(0.004)	
Panel B. NREGS per year, 2012–2014				
Households that asked for work	0.3	-0.5	1.8	87.8
	(1.6)	(1.9)	(1.4)	
Households employed	0.5	-0.2	1.9	79.6
	(1.5)	(1.8)	(1.3)	
Person-days	3.5	46.0	35.2	2,573.3
	(60.0)	(71.5)	(54.4)	
Joint test, p-value	0.700	0.340	0.247	
GPs	20,588	11,984	22,453	

Notes: Columns 1–3 report the difference in each characteristic between GPs just above and just below the cutoff, estimated within each group's main bandwidth (0.668 standard deviations for SC, 0.331 for OBC, and 0.318 for women's reservation), with a triangular kernel and block fixed effects. Column 4 reports the average across all GPs. Rows starting with "Villages with" show the fraction of a GP's villages that have that facility; for example, 0.812 in column 4 for primary schools means that, on average, 81 percent of a GP's villages have a primary school. Data are from the 2011 Census (Panel A) and NREGS records (Panel B), compiled by Chaturvedi, Das and Mahajan (2025); Census data are missing for about 5 percent of GPs. The joint test asks whether all the differences are zero together. Standard errors, clustered by block, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2. Effect of Reservation on the Number of Approved Works: Robustness to Block-Level Shocks ↩

	Approved works per year		
	(1)	(2)	(3)
Women-reserved	0.08	0.07	0.08
	(0.09)	(0.08)	(0.08)
OBC-reserved	-0.82***	-0.83***	-0.82***
	(0.09)	(0.08)	(0.09)
SC-reserved	-1.54***	-1.55***	-1.54***
	(0.10)	(0.09)	(0.10)
Mean, not reserved by caste	22.43	22.43	22.43
GP-terms	99,054	99,050	99,054
Time fixed effects	Term	Block × term	Term
Errors clustered by	GP	GP	Block

Notes: Each observation is a GP-term, and each column estimates equation (1) for the number of works approved per year. Column 1 includes GP and term fixed effects. Column 2 replaces the term fixed effects with block-by-term fixed effects, comparing each GP only with GPs in the same block and term; this drops two GPs that are the only GP in their block. Column 3 repeats column 1 with standard errors clustered by block. ST reservation is included as a control but not reported. The mean row reports the average for GP-terms not reserved by caste. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3. The SC Effect on Seats Also Reserved for Women ↩

	(1) Completed	(2) Abandoned
SC-reserved	-0.0085***	0.0061***
	(0.0016)	(0.0013)
Women-reserved	-0.0029**	0.0005
	(0.0012)	(0.0010)
SC-reserved × women-reserved	0.0024	0.0018
	(0.0028)	(0.0024)
Mean of outcome	0.698	0.196
Observations	10,149,181	10,149,181

Notes: The regression adds an interaction between SC and women's reservation to equation (2), using approved works. OBC and ST reservation are included as controls. Standard errors, in parentheses, are clustered by GP. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4. The SC Effect: Robustness to Block-Level Shocks ↩

	(1) Ongoing	(2) Completed	(3) Abandoned
Baseline	-0.0004	-0.0076***	0.0067***
	(0.0005)	(0.0012)	(0.0010)
Block × start-year fixed effects	-0.0002	-0.0073***	0.0062***
	(0.0004)	(0.0010)	(0.0008)
Standard errors clustered by block	-0.0004	-0.0076***	0.0067***
	(0.0004)	(0.0011)	(0.0009)
Mean of outcome	0.062	0.698	0.196
Works	10,149,181	10,149,181	10,149,181
Blocks	810	810	810

Notes: Each cell is the SC-reserved coefficient from a separate regression of the indicated outcome on the reservation indicators, using approved works. The first row is the baseline specification of equation (2), with GP and start-year fixed effects. The second row replaces the start-year fixed effects with block-by-start-year fixed effects, so that each work is compared only with works recorded in the same block and year. This drops two works that are alone in their block and year. The third row repeats the baseline with standard errors clustered by block. All regressions control for women's, OBC, and ST reservation. Standard errors, in parentheses, are clustered by GP in the first two rows and by block in the third. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A5. Effect on Abandonment for GPs That Gained and Lost Reservation ↩

	(1) SC	(2) OBC	(3) Women
Change after gaining reservation	0.014***	-0.003	0.003*
	(0.002)	(0.002)	(0.002)
Change after losing reservation	0.001	-0.010***	0.001
	(0.002)	(0.002)	(0.002)
Mean of outcome	0.196	0.196	0.197
Observations	10,147,871	10,145,502	9,565,657

Notes: The outcome is whether an approved work was abandoned. Each column covers one reservation group and compares GPs that gained reservation for that group in 2021 (not reserved in 2015, reserved in 2021) or lost it (reserved in 2015, not reserved in 2021) with GPs never reserved for the group. The first row is the change in abandonment between the two terms in GPs that gained reservation, minus the same change in never-reserved GPs. The second row is the same comparison for GPs that lost reservation. A positive coefficient means abandonment rose more, or fell less, than in never-reserved GPs. All regressions include GP and start-year fixed effects and control for the other reservation indicators. Standard errors, in parentheses, are clustered by GP. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6. Regression Discontinuity Estimates of the Number of Approved Works per Year ↩

	(1) MSE-optimal bandwidth	(2) Half the bandwidth	(3) Quarter of the bandwidth	(4) With extra controls
SC-reserved	-1.48***	-1.67***	-1.97***	-1.50***
	(0.33)	(0.46)	(0.64)	(0.33)
Percent of mean below cutoff	-6.4	-7.2	-8.5	-6.4
Mean below cutoff	23.21	23.21	23.33	23.30
GPs	20,584	10,654	5,569	19,600
OBC-reserved	-0.37	-0.60	0.24	-0.49
	(0.52)	(0.89)	(1.43)	(0.52)
Percent of mean below cutoff	-1.6	-2.6	1.1	-2.1
Mean below cutoff	23.04	23.00	23.06	23.07
GPs	11,980	6,513	3,538	11,429
Women-reserved	-0.79*	-0.68	-0.09	-0.87**
	(0.41)	(0.62)	(0.93)	(0.42)
Percent of mean below cutoff	-3.6	-3.1	-0.4	-4.0
Mean below cutoff	21.61	21.93	22.14	21.66
GPs	22,450	12,258	6,884	21,185

Notes: Each column reports fuzzy regression discontinuity estimates of the effect of reservation on the number of works approved per year in a GP during the 2015 council term. All regressions include block fixed effects, control for GP population, and weight each GP by the triangular kernel. Column 1 uses the MSE-optimal bandwidth (0.668 for SC, 0.331 for OBC, and 0.318 for women's reservation); columns 2 and 3 use half and a quarter of it. Column 4 adds the number of villages, GP area, and average NREGS person-days in 2012–2014, and drops GPs missing any of these. Standard errors, clustered by block, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A7. Regression Discontinuity Estimates without GPs Closest to the Cutoff ↩

	(1) Ongoing	(2) Completed	(3) Abandoned
Panel A. All GPs within the bandwidth			
SC-reserved	-0.0003	-0.0118***	0.0119***
	(0.0007)	(0.0034)	(0.0034)
GPs	20,584	20,584	20,584
OBC-reserved	0.0004	-0.0040	0.0034
	(0.0011)	(0.0050)	(0.0048)
GPs	11,978	11,978	11,978
Women-reserved	0.0007	-0.0057	0.0050
	(0.0008)	(0.0039)	(0.0037)
GPs	22,450	22,450	22,450
Panel B. GPs within 0.02 SD of the cutoff dropped			
SC-reserved	0.0003	-0.0108***	0.0102***
	(0.0008)	(0.0038)	(0.0037)
GPs	19,600	19,600	19,600
OBC-reserved	-0.0005	-0.0055	0.0055
	(0.0012)	(0.0055)	(0.0053)
GPs	10,826	10,826	10,826
Women-reserved	0.0004	-0.0071*	0.0068*
	(0.0009)	(0.0043)	(0.0041)
GPs	20,226	20,226	20,226
Panel C. GPs within 0.05 SD of the cutoff dropped			
SC-reserved	0.0004	-0.0100**	0.0094**
	(0.0009)	(0.0043)	(0.0042)
GPs	18,515	18,515	18,515
OBC-reserved	-0.0017	-0.0092	0.0105*
	(0.0015)	(0.0065)	(0.0063)
GPs	9,618	9,618	9,618
Women-reserved	0.0006	-0.0060	0.0056
	(0.0010)	(0.0055)	(0.0054)
GPs	17,665	17,665	17,665

Notes: The table checks whether the estimates depend on the GPs closest to the cutoff, where any sorting would show up. All estimates are fuzzy regression discontinuity estimates for the councils elected in 2015, using the category's MSE-optimal bandwidth (0.668 standard deviations for SC, 0.331 for OBC, and 0.318 for women's reservation), a triangular kernel, and block fixed effects, with each GP weighted by its number of approved works. The outcomes are the shares of a GP's approved works that are ongoing, completed, or abandoned. Panel A uses all GPs within the bandwidth. Panels B and C drop GPs within 0.02 and 0.05 standard deviations of the cutoff. Standard errors, clustered by block, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8. Regression Discontinuity Estimates without GPs Reserved for Women ↩

	(1) Reserved for women	(2) Ongoing	(3) Completed	(4) Abandoned	(5) Approved works per year
Panel A. All GPs					
SC-reserved	-0.212***	-0.0003	-0.0118***	0.0119***	-1.48***
	(0.011)	(0.0007)	(0.0034)	(0.0034)	(0.33)
Mean below the cutoff	0.241	0.017	0.780	0.202	23.21
GPs	20,584	20,584	20,584	20,584	20,584
OBC-reserved	-0.106***	0.0004	-0.0040	0.0034	-0.37
	(0.016)	(0.0011)	(0.0050)	(0.0048)	(0.52)
Mean below the cutoff	0.151	0.018	0.778	0.204	23.04
GPs	11,978	11,978	11,978	11,978	11,980
Panel B. Without women-reserved GPs					
SC-reserved		-0.0005	-0.0097***	0.0100***	-1.69***
		(0.0007)	(0.0036)	(0.0035)	(0.34)
Mean below the cutoff		0.018	0.776	0.206	22.07
GPs		17,052	17,052	17,052	17,052
OBC-reserved		0.0001	-0.0016	0.0013	-0.61
		(0.0011)	(0.0051)	(0.0049)	(0.51)
Mean below the cutoff		0.018	0.773	0.208	21.85
GPs		11,196	11,196	11,196	11,198

Notes: The table reports fuzzy regression discontinuity estimates for the councils elected in 2015, at the MSE-optimal bandwidth (0.668 standard deviations for SC and 0.331 for OBC reservation). Column 1 is whether the Sarpanch position was also reserved for women. Columns 2 to 4 are the shares of a GP's approved works that are ongoing, completed, or abandoned, and column 5 is the number of works approved per year in the GP. Columns 1 to 4 weight each GP by the triangular kernel times its number of approved works. Column 5 weights each GP by the kernel only, uses GPs with at least one work, and controls for GP population. All regressions include block fixed effects. Panel A uses all GPs within the bandwidth, and Panel B drops the GPs reserved for women. Means below the cutoff are weighted as in the regressions. Standard errors, clustered by block, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Supplementary Appendix: Data Construction

This section describes how I build the sample of public works from the NREGS administrative records and how I define the outcomes.

Work records and council terms

The work records come from the NREGS management information system (MIS), which contains every public work recorded for a GP since 2005, most of them under NREGS. For each work, I observe its code, status, type, wage and material payments, person-days generated, and start and completion dates. I collected the records for all GPs in Uttar Pradesh in August 2026, so each work's status is its status at that date.

I assign each work to a council term using the start date recorded in the MIS. For works that began, this is the date on which work started. For works that never began, the recorded date marks the financial year in which the work was approved and is most often 1 April, the first day of that year. In the sample, 55 percent of works marked Closed and 59 percent of works marked Approved are dated 1 April, compared with 0.3 percent of works marked Completed and 0.2 percent of works marked Ongoing.

The sample covers the 49,527 GPs that can be matched across the two elections and have works in both terms. I exclude 30 matched GPs with no works in at least one term: 23 have works in only one term, and 7 have none in either. A term without works leaves no record in the MIS, so for these GPs I cannot tell whether the council undertook no works or the records are missing. For consistency, the regression discontinuity analysis also excludes the 17 GPs with no works under the 2015 councils. The resulting matched dataset contains 10,964,069 work records, 5,838,266 under the councils elected in 2015 and 5,125,803 under those elected in 2021.

Work status

The MIS records one of eight statuses for each work: New, Approved, Ongoing, Suspended, Completed, Physically Completed, Closed, and Deleted.

Table S1. MIS Work Status and Outcome Categories ↩

MIS status	Meaning	Category	Works	Percent
Panel A. All work records				
New	Proposed but not yet approved	Excluded	68,766	0.6
Deleted	Record later removed from the system	Excluded	746,122	6.8
Statuses in Panel B	Approved for implementation	Approved works	10,149,181	92.6
Total			10,964,069	100.0
Panel B. Approved works				
Approved	Approved, but work has not begun	Not yet started	443,786	4.4
Ongoing	Work has begun and is not yet finished	Ongoing	620,965	6.1
Suspended	Work has begun but has been halted	Ongoing	5,546	0.1
Completed	Work finished and formally completed	Completed	6,969,500	68.7
Physically Completed	Work finished on the ground but not yet formally completed	Completed	115,547	1.1
Closed	Closed without being carried out	Abandoned	1,993,837	19.6
Total			10,149,181	100.0

Notes: The table covers NREGS works in the 49,527 GPs observed in both council terms, dated May 2016 to March 2021 under the councils elected in 2015 or November 2021 to March 2026 under the councils elected in 2021. Status is observed in August 2026. Panel A reports each status as a percentage of all work records, and Panel B as a percentage of approved works. Approved works are all works except those marked New or Deleted.

Two statuses fall outside the sample. Works marked New are proposals that have not yet been approved. They account for 0.6 percent of all records and belong almost entirely to the 2021 councils. Works marked Deleted are records later removed from the system, possibly because they were duplicates or entered by mistake. They account for 6.8 percent of all records, and 97 percent of them record no payment to workers or vendors. I exclude both groups, since new works have not yet been approved and deleted records cannot be assigned an implementation outcome.

The remaining six statuses make up the sample of approved works: 10,149,181 works, or 92.6 percent of all records. Approval makes a work eligible for implementation but does not ensure that it will be started or completed.

Outcomes

I group the six statuses of approved works into four categories. Every work in the sample has been approved, but in the MIS the status Approved has a narrower meaning: it marks works that were approved but on which no work had begun by August 2026. These works remain open and may still begin. To avoid confusion with the full sample of approved works, I call them *not yet started*. The second category is *ongoing* works, marked Ongoing or Suspended in the MIS. Work on these has begun but is not finished, and on suspended works it has been halted.

The third category is *completed* works, marked Completed or Physically Completed in the MIS. Works marked Completed are finished both on the ground and in the records. Works marked Physically Completed are finished on the ground, but their final bills or payments have not yet been settled. The fourth category is *abandoned* works, marked Closed in the MIS: they were closed without being carried out. Of the nearly 2 million abandoned works, 99.99 percent record no payment to workers or vendors, so they were not completed.

To measure how many works a council undertakes, I count each GP's approved works per year of the council term. To measure how those works are implemented, I treat each approved work w as an observation and use three binary outcomes:

$$\begin{aligned}
\text{Ongoing}_w &= \mathbf{1}\{\text{status}_w \in \{\text{Ongoing}, \text{Suspended}\}\}, \\
\text{Completed}_w &= \mathbf{1}\{\text{status}_w \in \{\text{Completed}, \text{Physically Completed}\}\}, \\
\text{Abandoned}_w &= \mathbf{1}\{\text{status}_w = \text{Closed}\}.
\end{aligned}$$

Works not yet started take the value zero on all three outcomes. I keep them in the sample, so the mean of each outcome is its share of all approved works, and the three shares and the share not yet started add up to one. Whether a work has started may itself depend on who leads the council, so dropping these works would compare different sets of works across reserved and unreserved terms.